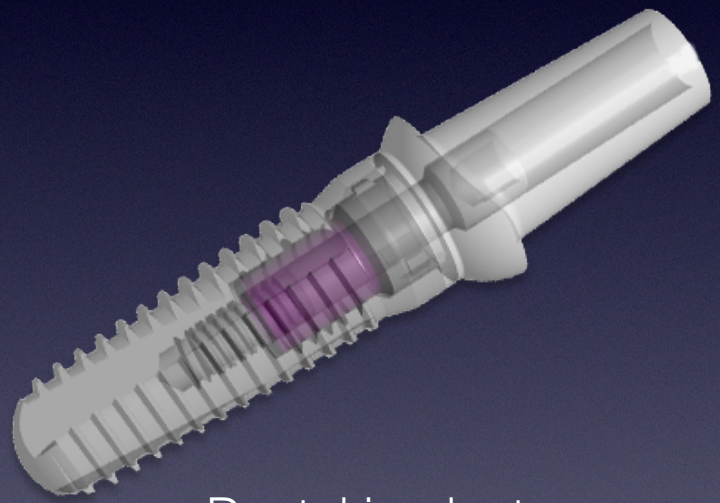
Touching 100.000.000 lives

Pork knuckle can be hard

Delicious and really crisp

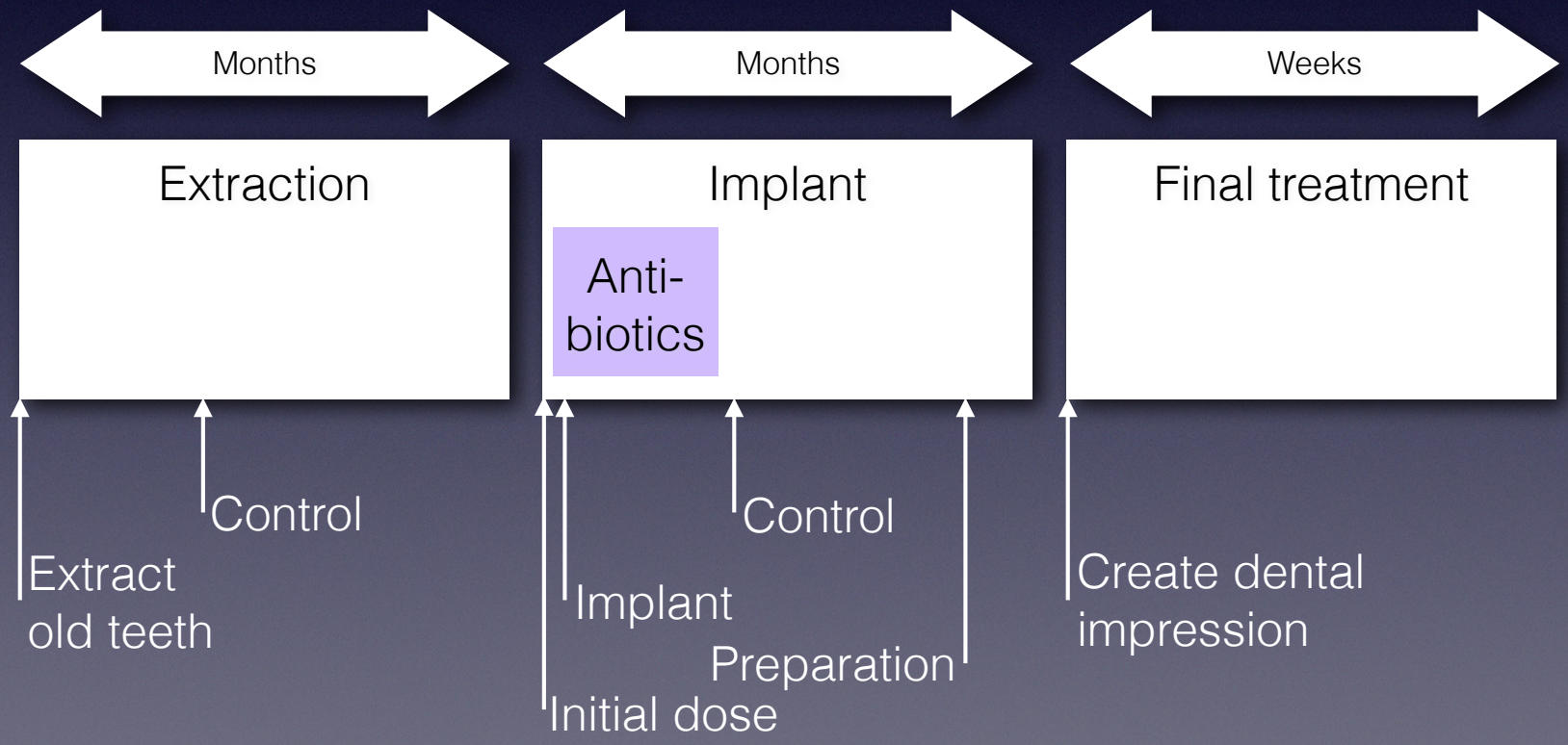
Sometimes too hard for teeth

The final result



Dental implant

To get it fixed can take some time



This is what I was told

Before surgery

Please take the antibiotics one hour before the surgery

Your next appointment

April 24
8am

After surgery

Please continue to take the full dose of the antibiotics

Some more

Please do not eat dairy products for next two days

What if there would be a way of conveying these message digitally?

Could the message be converted in script?

```
"dental implant" is a plan (
 "implant procedure" on 24/Apr/2014 at 08:00am.

 one hour before implant procedure (
  message to patient "Please take initial dose of your antibiotics".
 ).

 for 9 days after implant procedure (
  before breakfast, before dinner (
   message to patient "Please take your dose of antibiotics now".
  ).
 ).

 "intense aftercare" for 2 days after implant procedure (
  every morning (
   message to patient "Please do not eat dairy products".
  ).
 ).
 after intense aftercare (
  once before breakfast (
   message to patient "You can eat as you are used to." (
    message expires after 1 day.
   ).
  ).
 ).
).
```

A simple script describing the "journey"

Using a language with powerful features

Natural language

Secure to use

Simple to learn

Open

Safe to use

Easy to integrate

Focus on medical purpose

Local language support

Reflection of real world

Support the local nature of healthcare

```
"dental implant" is a plan (
 "implant procedure" on 24/Apr/2014 at 08:00am.

 one hour before implant procedure (
   show message to patient "Please take initial dose of your antibiotics".
 ).

 for 9 days after implant procedure (
   before breakfast, before dinner (
     show message to patient "Please take your dose of antibiotics now".
   ).
 ).

 "intense aftercare" for 2 days after implant procedure (
   every morning (
     show message to patient "Please do not eat dairy products".
   ).
 ).
 after intense aftercare (
   once before breakfast (
     show message to patient "You can eat as you are used to." (
       message expires after 1 day.
     ).
   ).
 ).
).
```

```
"Implantation" ist ein Plan (
 "Implantation" am 24/Apr/2014 um 08:00.

 Eine Stunde vor der Implantation (
   Mitteilung an den Patienten "Bitte nehmen Sie eine
     Antibiotika Tablette".
 ).

 Für 9 Tage nach der Implantation (
   Vor dem Frühstück, vor dem Abendessen (
     Mitteilung an den Patienten "Bitte nehmen Sie ihr
       Antibiotikum jetzt ein".
   ).
 ).

 "Intensive Nachsorge" für 2 Tage nach der Implantation (
   Jeden Morgen (
     Mitteilung an den Patienten "Bitte keine Milchprodukte essen".
   ).
 ).
 Nach der intensiven Nachsorge (
   Einmal vor dem Frühstück (
     Mitteilung an den Patienten "Sie können wieder wie gewohnt
     essen." (
       Mitteilung verfällt nach einem Tag.
     ).
   ).
 ).
).
```

The plan shown on this side is not medically validated (especially the medical decision) and is intended for illustrative purposes only.

Enabling creation of simple reminders to...

```
"simple" is a plan (
30 minutes before breakfast, 30 minutes before dinner (
  show message to patient "Please take your dose now" (
    message expires after 30 minutes.
  ).
).
).
```

A simple daily reminder

The plan shown on this side is not medically validated (especially the medical decision) and is intended for illustrative purposes only.

...to complex plans for chronic patients

```
"best inhalation period" is a period (
  between (Monday ... Friday) between (7:40 ... 7:50).
  between (Saturday ... Sunday) between (09:00 ... 09:10).
  each occurrence of period can be customized.
).

"fill in evening report" is a period (
  between (Monday ... Friday) between (18:30 ... 18:45).
  between (Saturday ... Sunday) between (19:45 ... 20:00).
  each occurrence of period can be customized.
).

"severity of breathlessness" is a value (
  value is valid if all of the following criteria apply (
    value is between(1 ... 5).
  ).
  meaning of value is (
    1 (meaning is "not at all". best value.).
    2 (meaning is "mild".).
    3 (meaning is "moderate".).
    4 (meaning is "severe".).
    5 (meaning is "worst possible". worst value.).
  ).
  question to capture value is "How severe was your breath
    validate answer once.
  ).

"use of rescue medication" is a number (
  value is valid if all of the following criteria apply (
    value is between(0 ... 50).
  ).
  question to capture value is "How severe was your breat
    validate answer once.
  ).
).

//-----------------------------------------------------------
// Stores when the inhaler was used.
//-----------------------------------------------------------
"inhaler used" is a discrete value (
  value is valid if all of the following criteria apply (
    value is one of ("Yes"; "No").
  ).
).
```

```
(//-----------------------------------------------------------
// Stores how often the inhaler was used in past 5 days, even it was outsi
//-----------------------------------------------------------
"inhaler used in past 5 days" is a number (
  value is valid if all of the following criteria apply (
    value is between (0 ... 5)?
  ).

  meaning of value is (
    0 (meaning is "all doses missed". worst value.).
    1 (meaning is "most doses missed".).
    2 (meaning is "almost all doses missed".).
    3 (meaning is "many doses missed".).
    4 (meaning is "most doses taken".).
    5 (meaning is "all doses taken". best value.).
  ).

  calculation of value is (
    timeframe for analysis is past 5 days (
      value is count of inhaler used is yes.
    ).
  ).

  presentation of value is (
    if value is 0 then (
      presentation of value (
        to patient is "Inhaler not used
        to physician is "Patient is not
      ).
    )
    if value is 1 then (
      presentation of value (
        to patient is "Inhaler usage *".
        to physician "Patient used the inhaler 1 time in past 5 days".
      ).
    ).
    if value is 2 then (
      presentation of value (
        to patient is "Inhaler usage **".
        to physician "Patient used the inhaler 2 times in past 5 days".
      ).
    ).
    if value is 3 then (
      presentation of value (
        to patient is "Inhaler usage ***".
        to physician "Patient used the inhaler 3 times in past 5 days"
      ).
    ).
```

```
    if value is 4 then (
      presentation of value (
        to patient is "Inhaler usage ****".
        to physician "Patient used the inhaler 4 times in past 5 days".
      ).
    ).
    if value is 5 then (
      presentation of value (
        to patient is "Inhaler usage *****".
        to physician "Patient used the inhaler 5 times in past 5 days".
      ).
    ).
  ).
).

//-----------------------------------------------------------
// Stores how often the inhaler was used in past 5 days on time.
//-----------------------------------------------------------
"inhaler used in past 5 days on time" is a value (
  value is valid if all of the following criteria apply (
    value is between (0 ... 5)?
  ).

  meaning of value is (
    0 (meaning is "
```

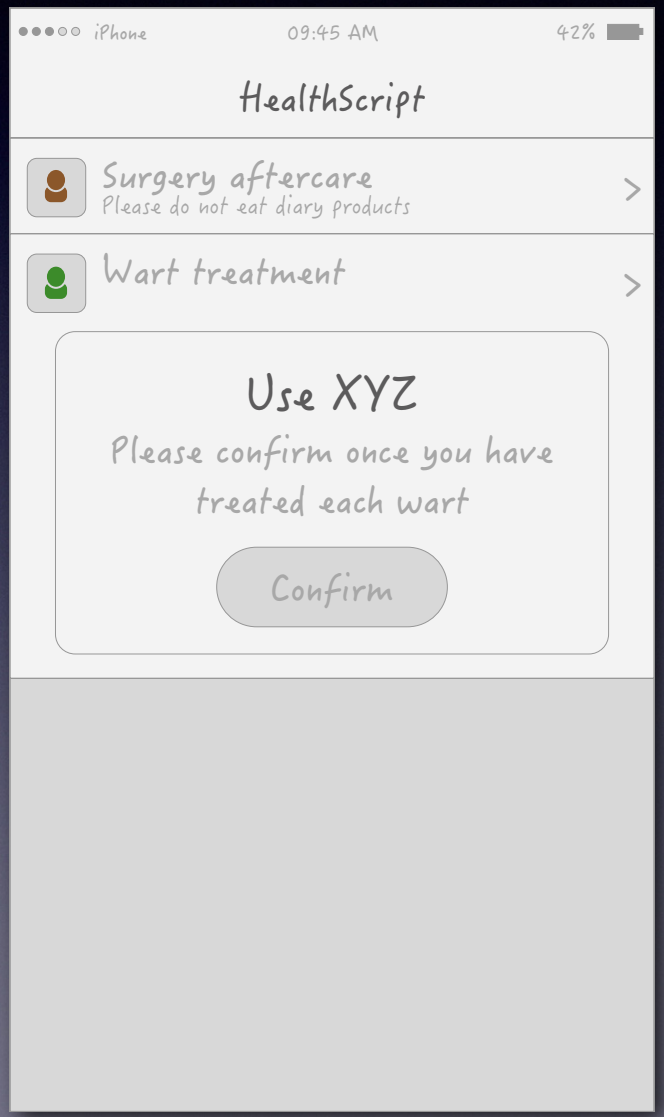
```
    timeframe for analysis is past 5 days within best inhalation period
      value is count of inhaler used is yes.
    ).
  ).
).

//-----------------------------------------------------------
// The full plan
//-----------------------------------------------------------
"COPD telehealth" is a plan (

  //---------------------------------------------------------
  // At the beginning of the inhalation period we show a reminder an
  // of use. If no answer is given until the event happens next time, w
  // actions.
  //---------------------------------------------------------
  "inhalation" with begin of best inhalation period (
    in case actions are not finished until next inhalation abort outstan
    message "Take now".
    ask for inhaler used.
  ).
```

```
  //---------------------------------------------------------
  // Ask for the daily reporting. If the report is not filled in until the next report is due we abort
  // the outstanding questions.
  //---------------------------------------------------------
  "reporting" with begin of fill in evening report (
    in case actions are not finished until next reporting abort outstanding actions.
    ask for severity of breathlessness.
    ask for use of rescue medication.

    // Check if the use of rescue medication is increasing and issue a warning if this is the case.
    timeframe for analysis is past 5 days (
      if trend of use of rescue medication increases then (
        message "You should come and see your physician".
      ).
      in all other cases of trend of use of rescue medication (
        no action.
      ).
    ).
  ).

  //---------------------------------------------------------
  // Every day after breakfast we provide a feedback based on a rolling review of past 5 days.
  //---------------------------------------------------------
  very day after breakfast (
  // daily feedback.
  f inhaler used in past 5 days on time is one of (most doses on time; all doses on time) then
    message "You are doing a great job in using the inhaler right on time".

  n all other cases of inhaler used in past 5 days on time (
    if inhaler used in past 5 days is 0 (
      // The inhaler is not used.
      message "Not using a medication does not lead to improvement!".
    ).
    if inhaler used in past 5 days is between (1 ... 2) (
      // The inhaler is not used.
      message "You must use the inhaler more regular".
    ).
    in all other cases of inhaler used in past 5 days (
      // The inhaler was used, but not at the intended time.
      message "You are using the inhaler frequently enough, but you should also use at the
        same time every day."
    ).
  ).
).
```

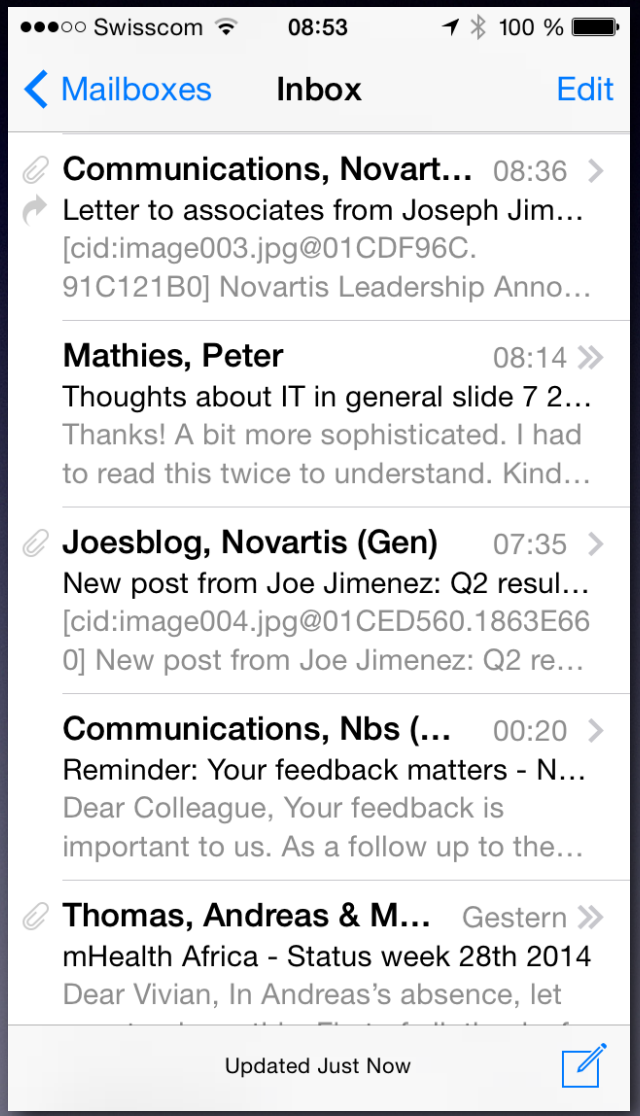
Managing a COPD patient

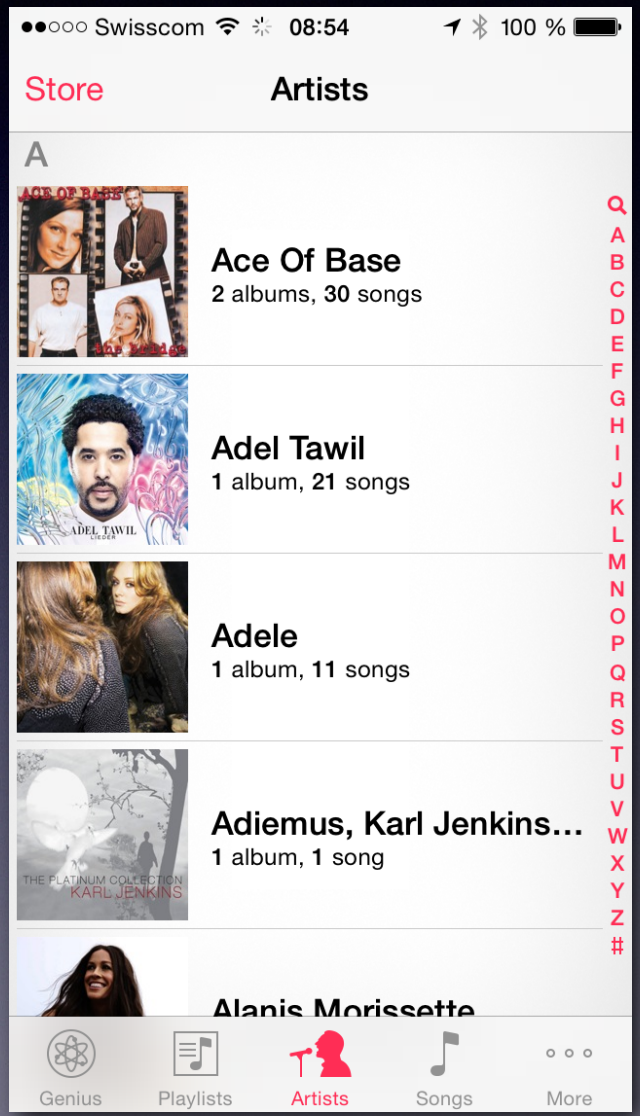
The plan shown on this side is not medically validated (especially the medical decision) and is intended for illustrative purposes only.

Enable one place for health...



...like you have for mail and music

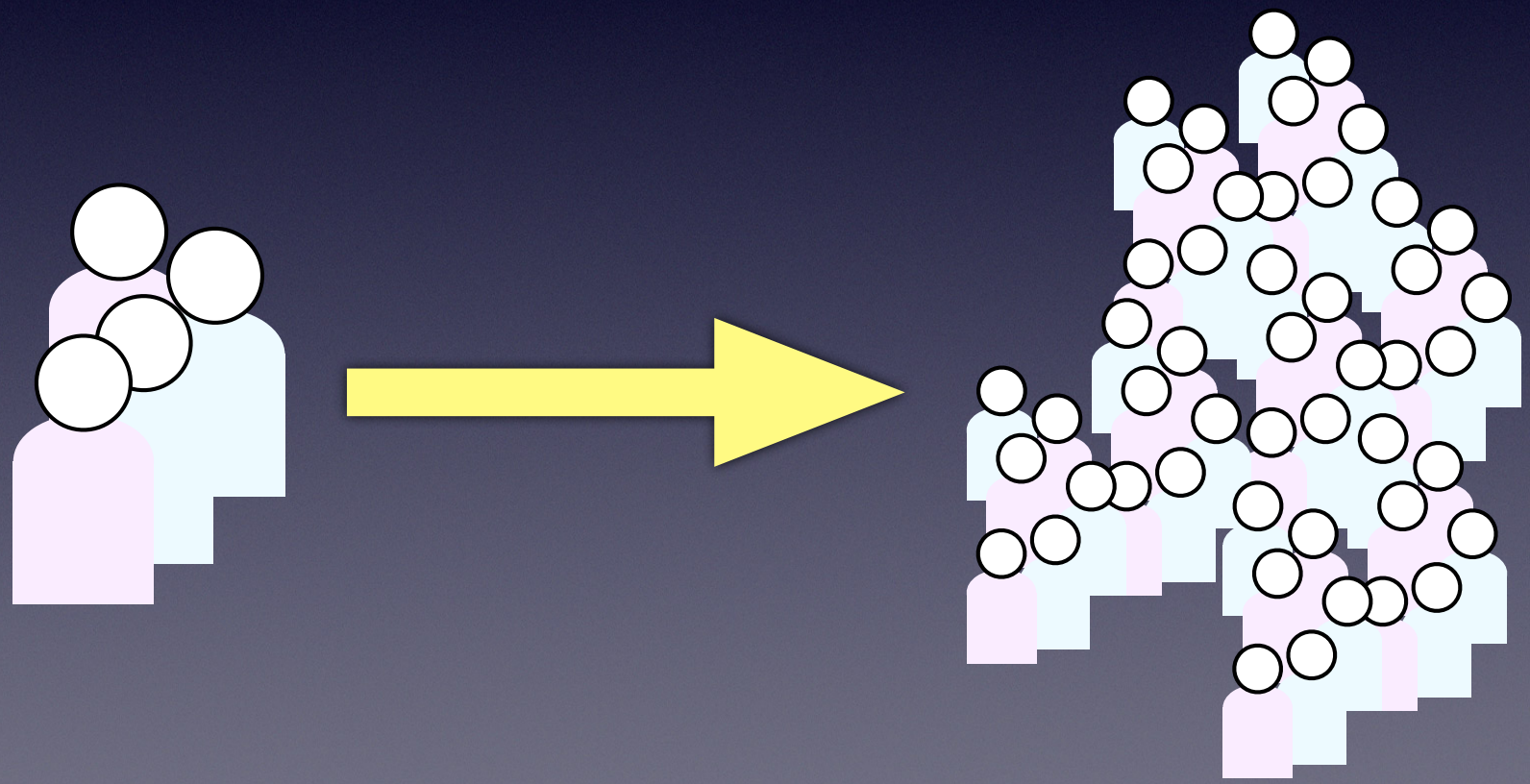




Finally Hippocrates can touch millions of lives

Thousands

Hundreds of millions

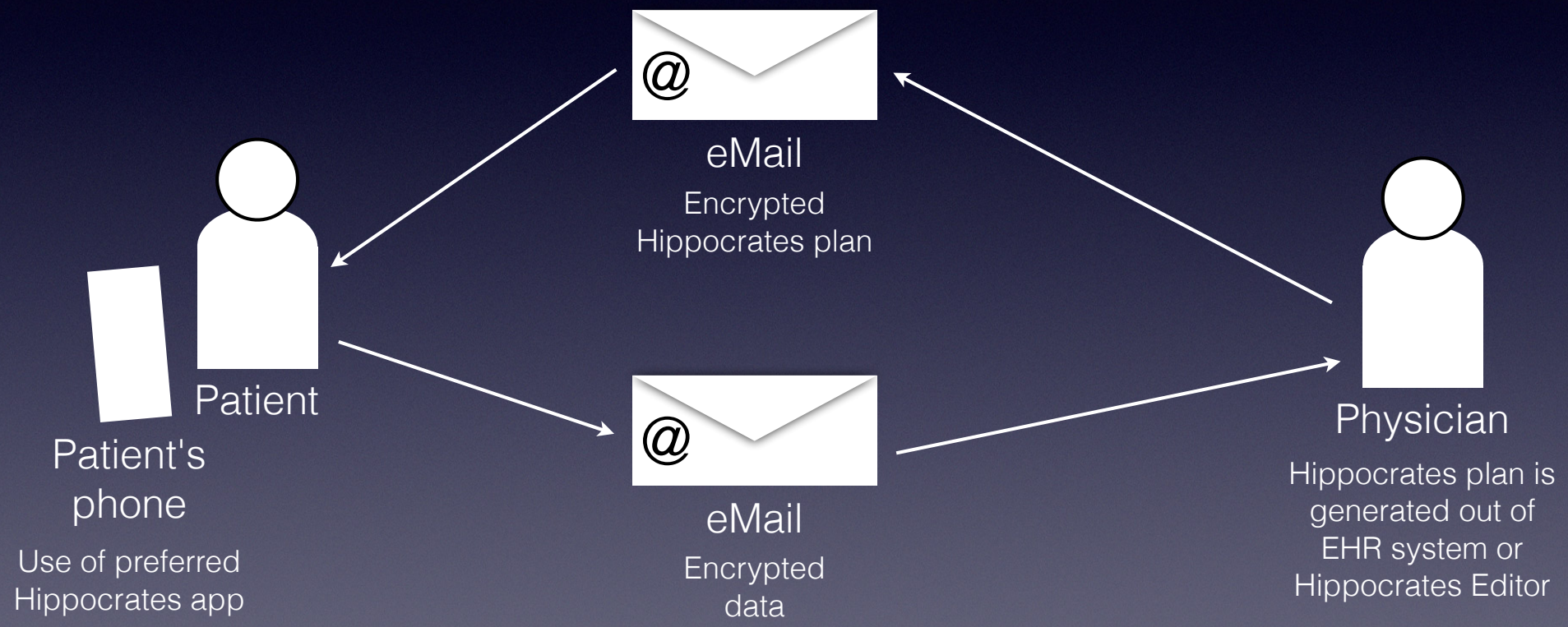


That is Hippocrates

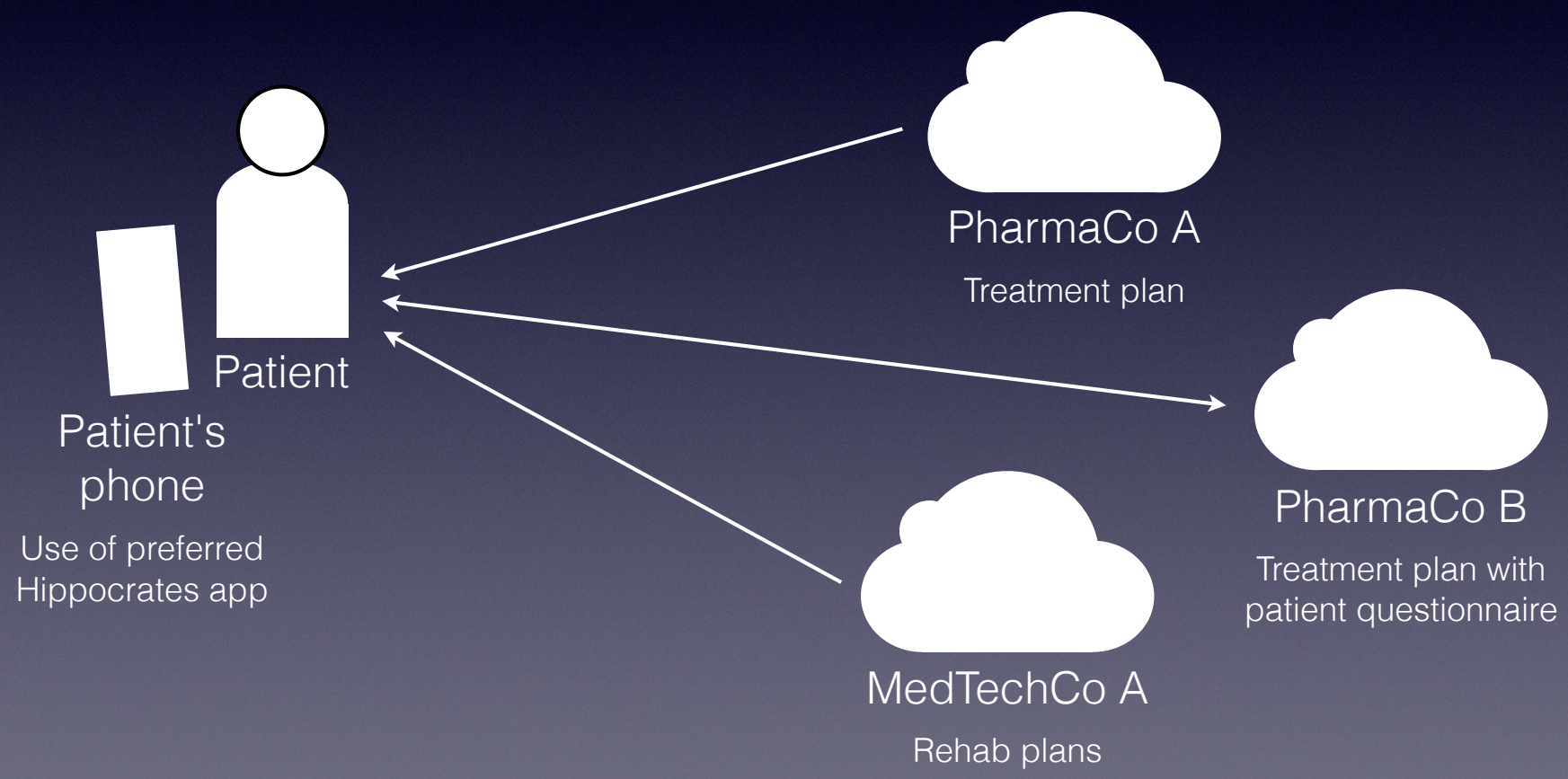
- **SAFE & EASY** - Hippocrates is a versatile language specifically designed for use in healthcare
- **FAST** - Hippocrates enables any stakeholder to provide better personalized care and communication within minutes not months
- **MOBILE** - Hippocrates delivers true mobility available anytime anytime

How to use Hippocrates?

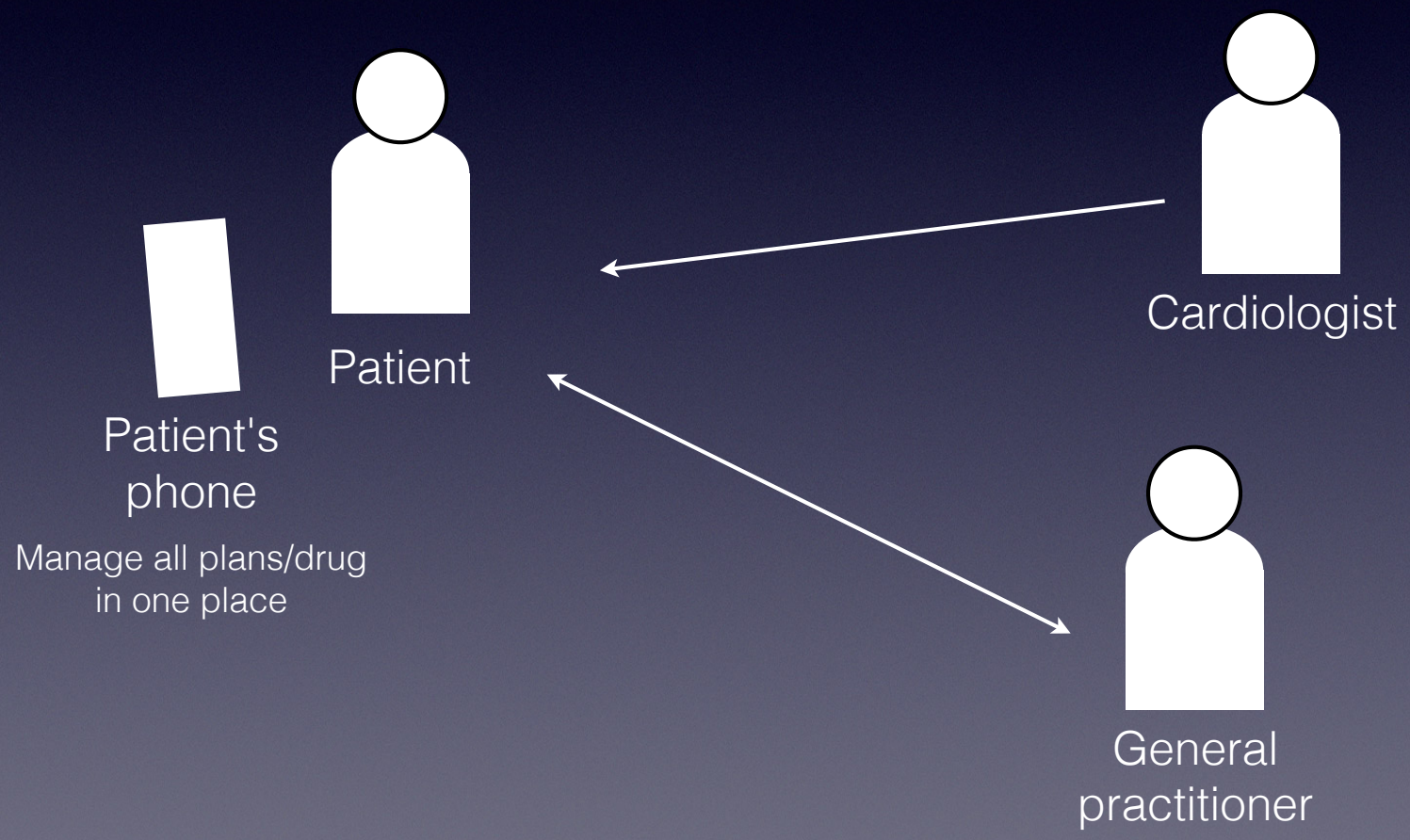
Communication without cloud back-end possible



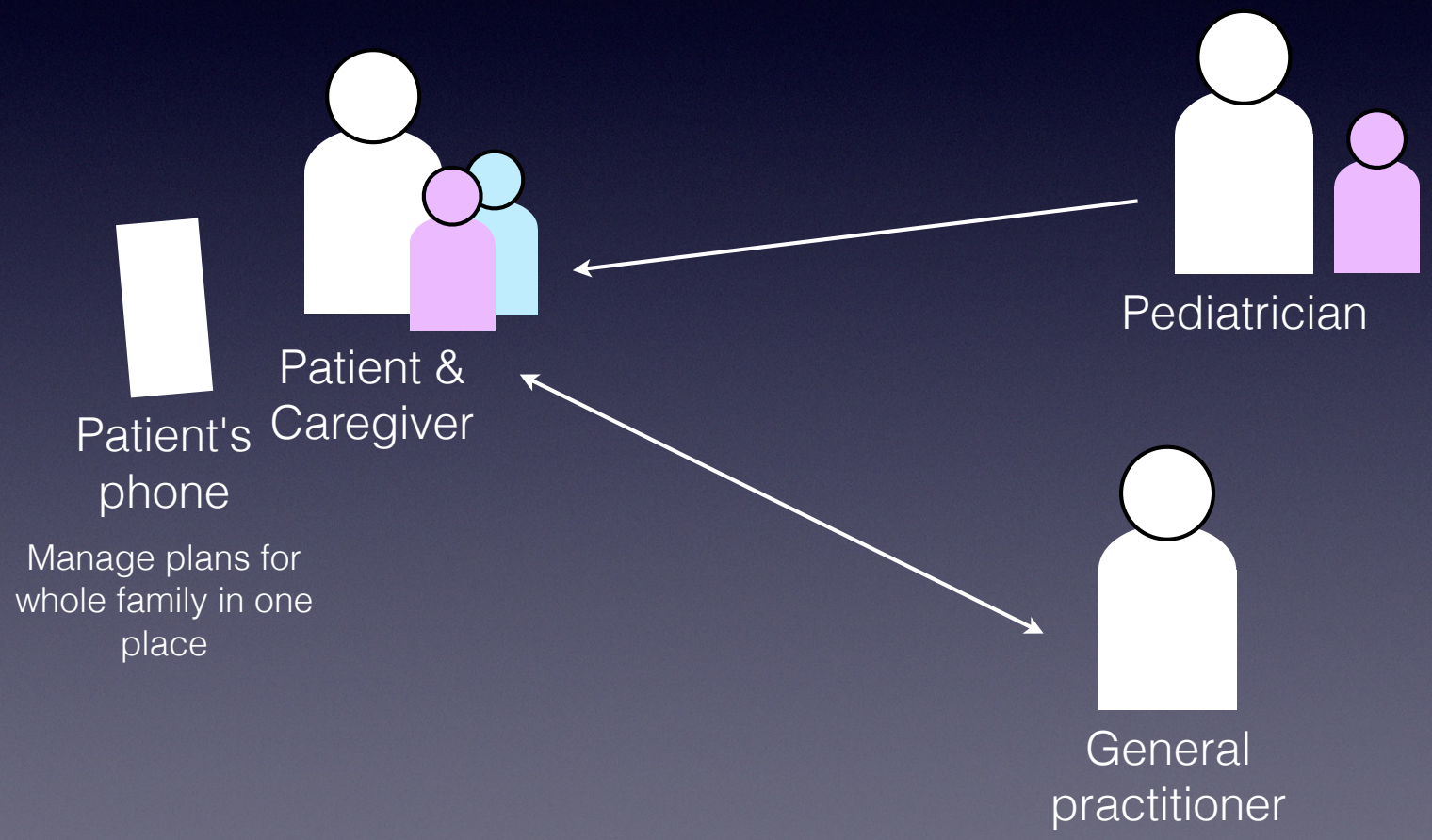
Easy access to better treatment



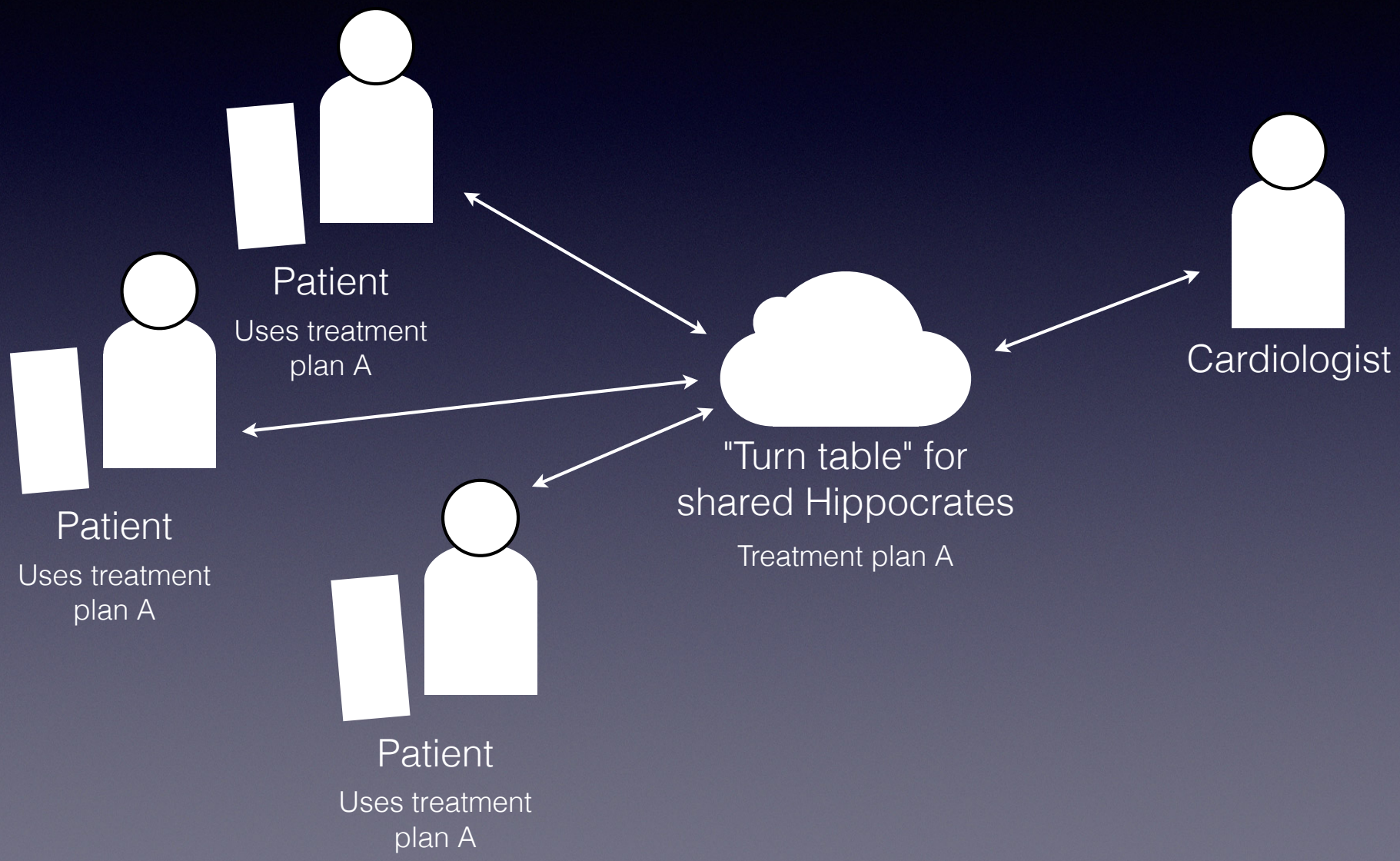
One place


Patient
Cardiologist
Patient's phone
Manage all plans/drug in one place
General practitioner

Even one place for the whole family

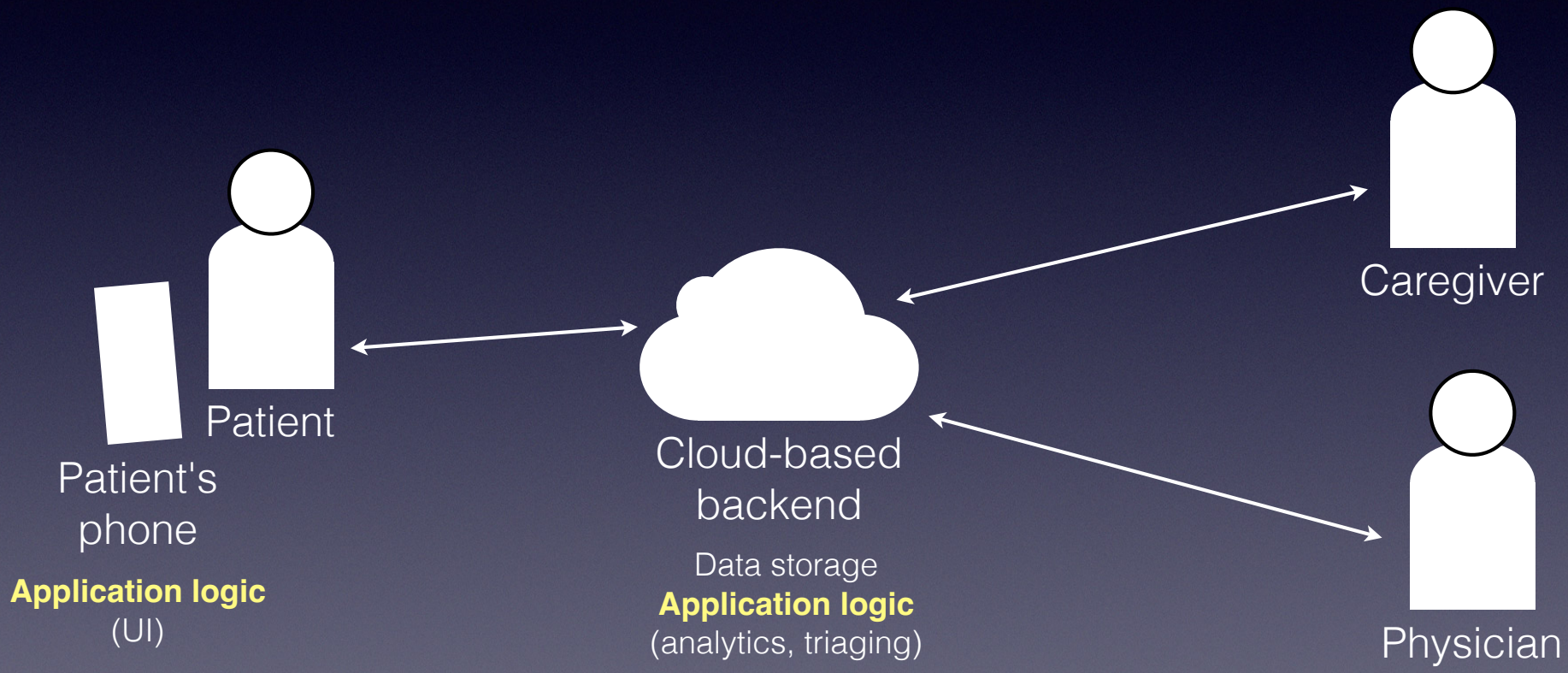


HealthScrips can easily also be shared

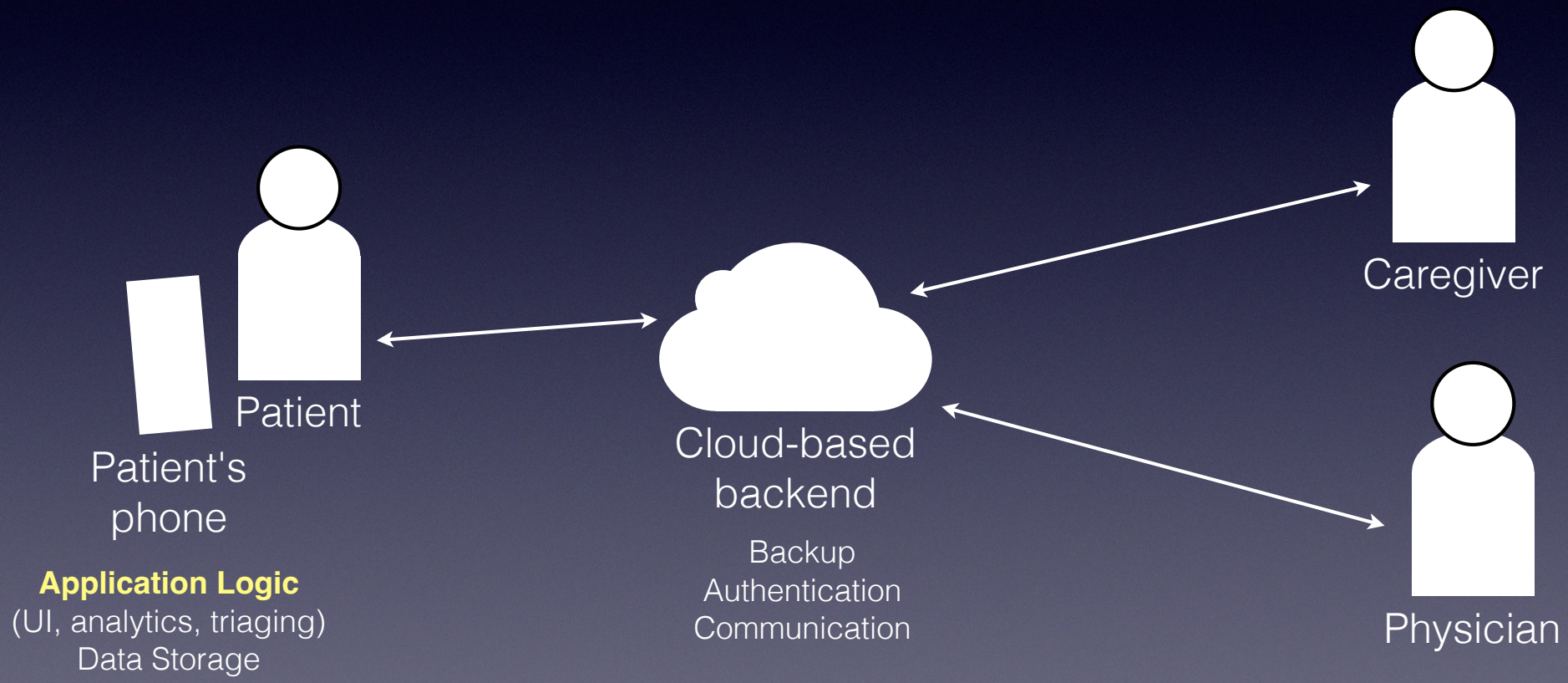


The Hippocrates Architecture

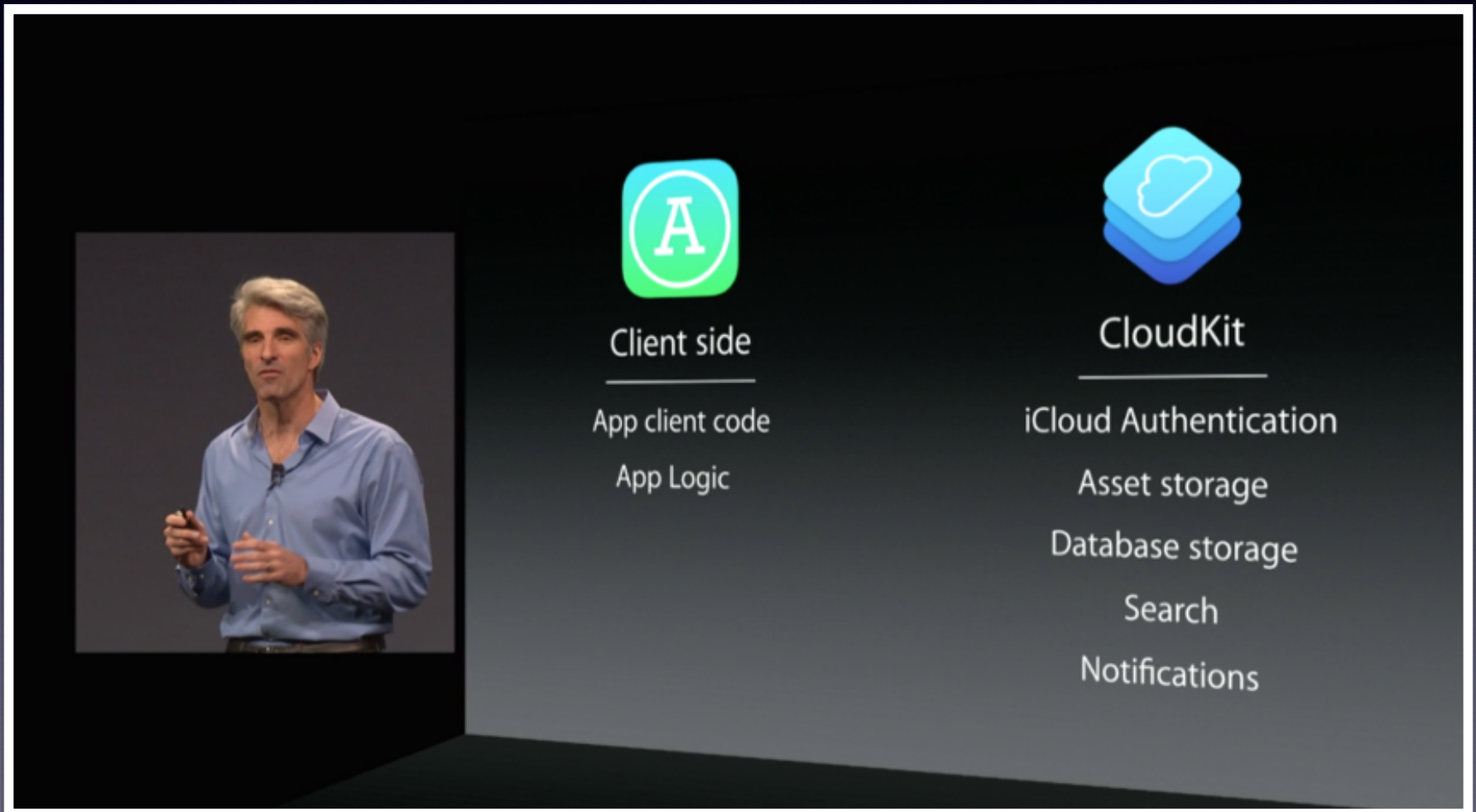
Typical architecture splits application logic



Hippocrates bundles all logic in one place



These changes are also proposed by Apple



Reduce time for changes from months to minutes

Typical architecture

1.	Specify requirements	**Weeks**
2.	Change mobile applications	**Weeks**
3.	Change cloud-backend	**Months**
4.	Deploy mobile applications to app-stores	**Weeks**
5.	Deploy modified cloud-backend	**Weeks**
		Several Months

Hippocrates based architecture

1.	Change script	**Minutes**
2.	Send to patient	**Minutes**
		Some Minutes

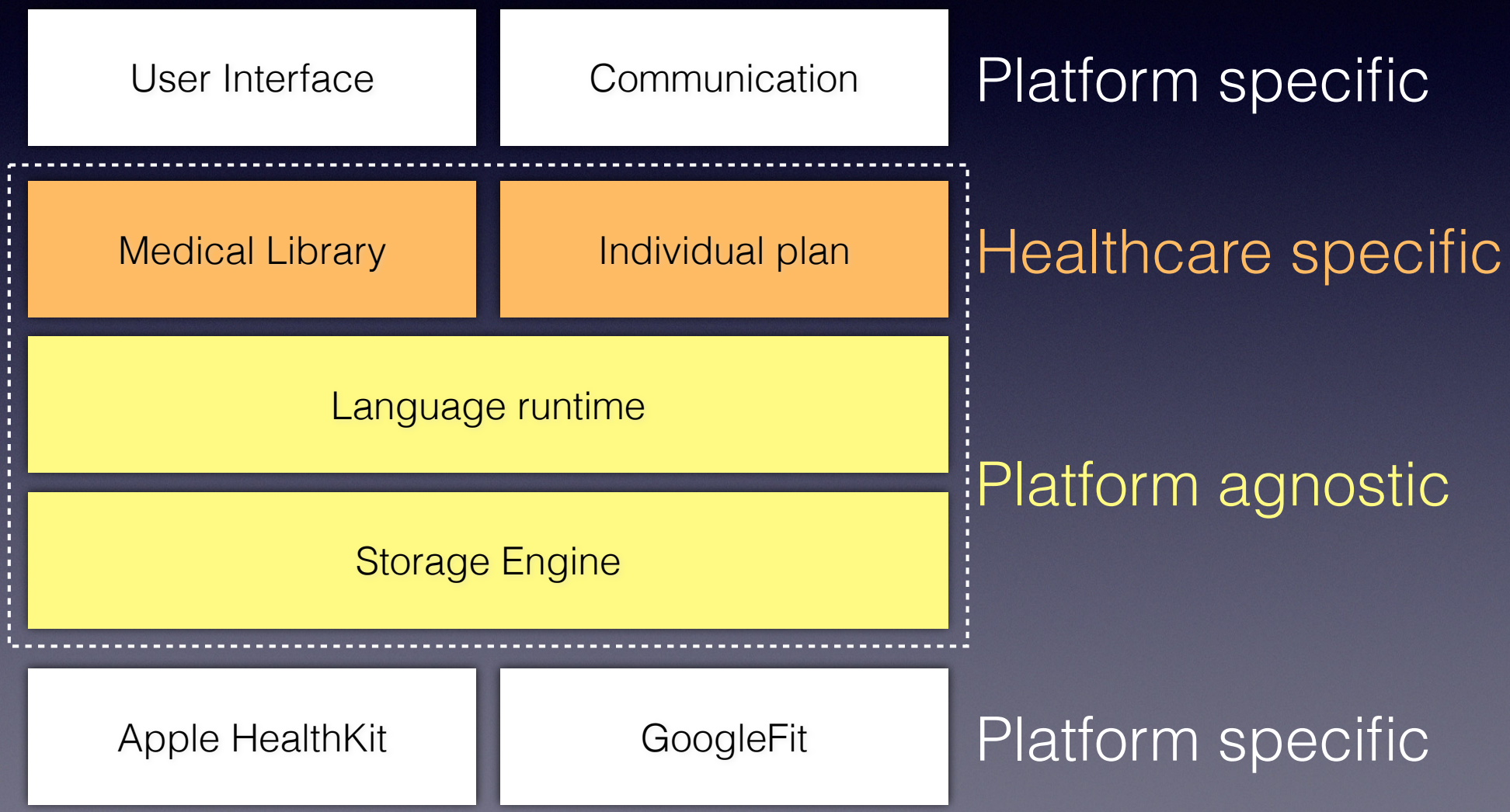
Hippocrates does not require any change to underlaying IT architecture

Change of architecture comes with benefits

- If the cloud fails, full logic is available on the phone close the patient
- Personal health data is protected on a personal device and does not travel, although it can be shared on request (*)
- Simplified and accelerated deployment

(*) If the phone fails/terminal there is also no data coming to a cloud based decision system

Embeddable architecture

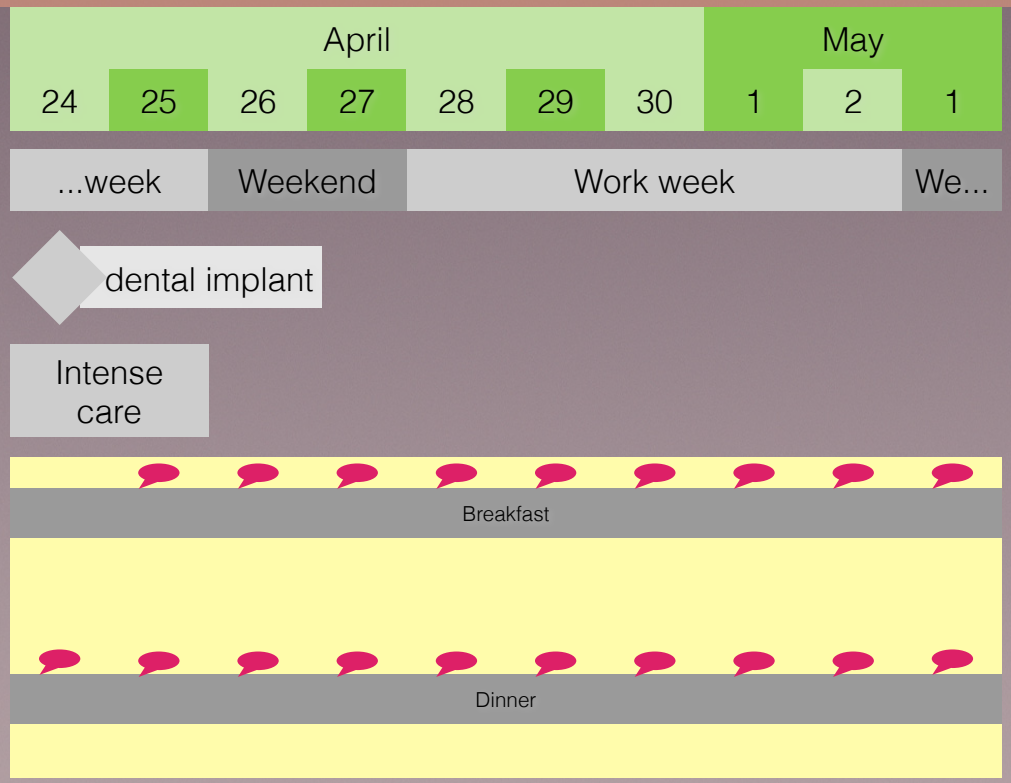


Hippocrates Workbench

Script

```
"dental implant" is a plan (
 "implant procedure" on 24/Apr/2014 at 08:00am.

 one hour before implant procedure (
   message to patient "Please take initial dose of your antibiotics".
 ).

 for 9 days after implant procedure (
   before breakfast, before dinner (
     message to patient "Please take your dose of antibiotics now".
   ).
 ).

 "intense aftercare" for 2 days after implant procedure (
   every morning (
     message to patient "Please do not eat dairy products".
   ).
 ).
 after intense aftercare (
   once before breakfast (
     message to patient "You can eat as you are used to." (
       message expires after 1 day.
     ).
   ).
 ).
).
```

Preview of "dental implant"



The plan shown on this side is not medically validated (especially the medical decision) and is intended for illustrative purposes only.

Hippocrates separates medical and technical domain

Medical

Medical knowledge

Treatment plans

Hippocrates runtime

Technical

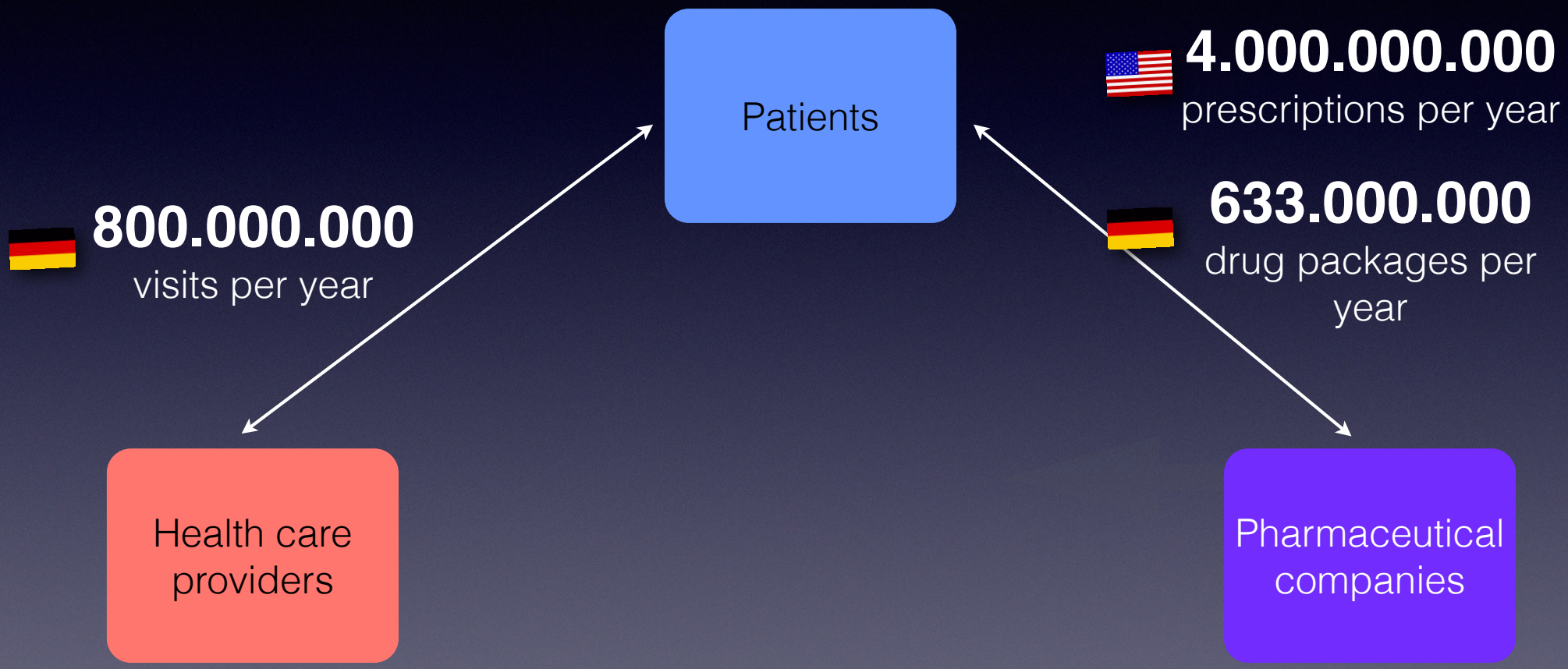
Embedding of runtime in host environments

Visualization of data

Communication with the "world"

Let's talk about business

Billions of use cases every year



Many benefits

Pharmacies

- Better customer service & support

Patients

- Better care & outcomes
- Simplified access

Payers

- Better care & outcomes for their customers
- Tracking of real-world outcomes

Health care providers

- Better care & outcomes
- Improved process efficiency

Hippocrates eco-system

Pharmaceutical companies

- Better outcomes
- Access to real-world data

Each stakeholder can contribute

Pharmacies

- Create plans for their customers

Patients

- Use plans
- Share data

Payers

- Provide plans
- Provide patient apps

Health care providers

- Provide plans
- Share plans
- Access patient data

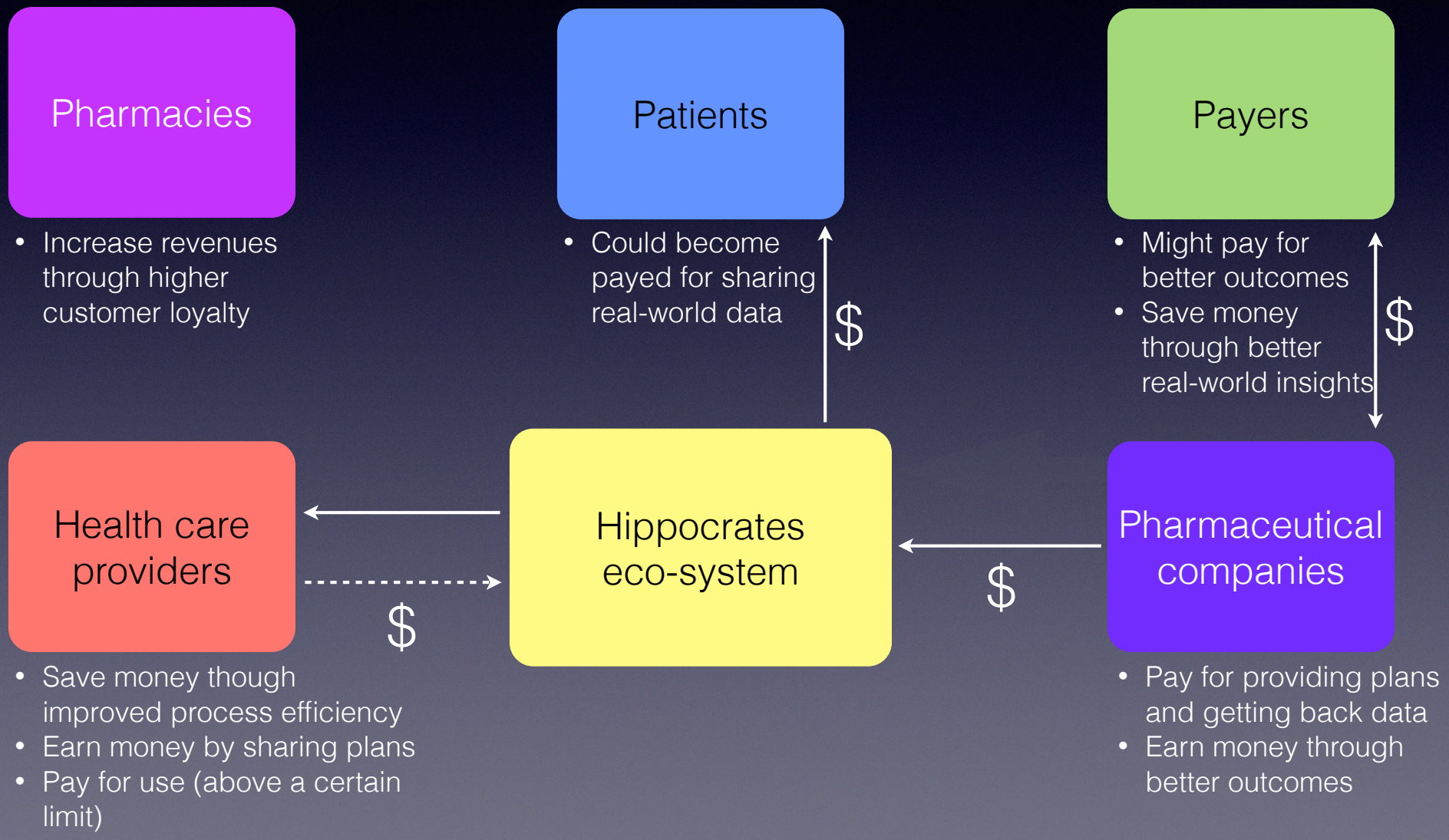
Hippocrates eco-system

- Provide platform for secure and compliant interaction

Pharmaceutical companies

- Provide plans
- Access data

Money flow in the Hippocrates eco-system



Different licensing models

First idea

	Pharmaceutical companies	Pharmacies Health care providers	Payers	Patients (application developers*)
Metric driven license	Volume of e.g. messages, questions & values	Volume of e.g. messages, questions & values	Active users or number of insured persons	Share of revenue
Base license	Base fee	Free	Free	Free

* Application developer in this case means a person or company outside a pharmaceutical company, a payer or a healthcare provider.

Metrics based licensing to pharmaceutical companies

Pharmaceutical companies

First idea

Consumption based license	Additional licenses based on usage points. A usage point can either be the duration of a plan or the number of interactions (questions & messages)
Base license	License for general use of Hippocrates.

Metrics based licensing to healthcare professionals

Healthcare professionals

First idea

Consumption based license	Additional licenses based on usage points. A usage point can either be the duration of a plan or the number of interactions (questions & messages) *
Base license	Free of charge up to a certain volume.

* This logic also applies if Hippocrates scrips are sold through healthcare professionals

Provision based licensing to EHR providers

EHR providers

First idea

Consumption based license	EHR provider get revenue share of licenses coming from healthcare professionals if scripts are generated with their systems.
Base license	Free

User based licensing to payers

Payers

First idea

User based license	License is based on peak active users per year.
Base license	Free of charge up to a certain number of users (e.g. 1 - 5% of all insured persons).

Revenue based licensing for application developers

Application developers*

First idea

Revenue based license	In case products that include the Hippocrates language are sold the license fee will be X% of revenue (including advertising and sale of whole application).
Base license	Free of charge.

* Application developer in this case means a person or company outside a pharmaceutical company, a payer or a healthcare provider.

Metrics based licensing for pharmacies

Pharmacies

First idea

Consumption based license	License will be based on interaction points
Base license	Free of charge.

Reference costs in diabetes care

Mobile Phone Diabetes Project 6 months*

Project lead to total average savings per patient within that 6 months period of 437 USD (or 8.8% savings over pre-period costs).

150 USD total technology costs** for 6 months

=

3.4 text messages to patient per day x 6 months
=
612 messages

=

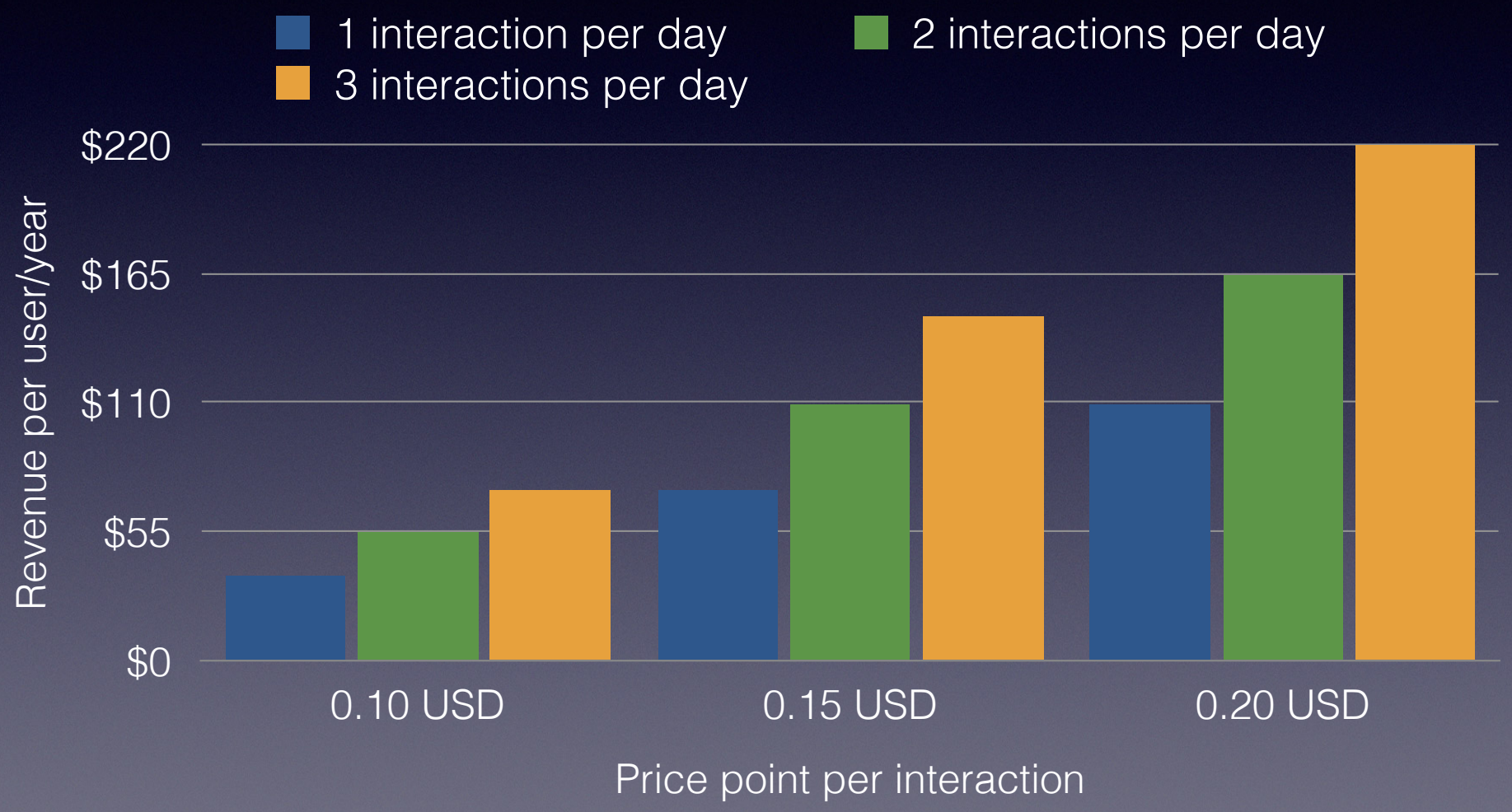
0.24 USD per message

* http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4034376/.
** Technology costs might include other technology related costs.

Intensity of interaction determines costs

	Simple plan	Complex plan
	Support of a longer term once per week treatment over a period of 8 weeks.	Support of a chronic patient with intense feedback over a period of 8 weeks.
Level of interaction	**1 message per week**	**5 messages** and **3 questions per day**
Total costs (@0.1 USD per interaction)	**0.8 USD**	**44.8 USD**

Revenue user/year



100.000.000 USD revenue
10.000.000 users

Eco-system

Credibility

Academia

Dublin
Cardiff

Users

Product

PracticeFusion
HealthTap
ZocDoc
Novartis
Teleadoc
eClinicalWorks
CareCloud

Speed of implementation

Implementation

Accenture
E&Y

Roadmap to implementation

Financing of early-stage required

better
4 - 5 mEUR

2 - 3 mEUR

Seed

- First draft of specification
- Definition of reference cases
- Business model
- First academic partners identified
- First regulatory feedback?

Early stage
(Series A)

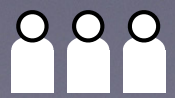
- First version of specification
- Beta version of Hippocrates environment (language core, mobile reference application and Hippocrates workbench)
- Implemented use cases
- Active academic community
- Business partnerships established (ideal pharmaceutical companies providers and EHR vendors)
- Regulatory opinion

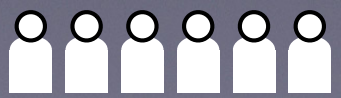
Draft version of specification in progress

- A first draft of language is in progress
- Most elements are specified or described

Full beta version can be available within 12 months

3 - 4 months	9 - 12 months	
Prototype	Beta version	Hippocrates environment
Hippocrates specification 0.5	Hippocrates specification 1.0	Hippocrates specification 1.0
Core language engine (selected features)	Core language engine (full feature set)	Core language engine (full feature set)
Reference mobile application	Reference mobile application	Reference mobile application
	Hippocrates Workbench	Hippocrates Workbench
	Reference medical library	Medical library
		EHR reference integration

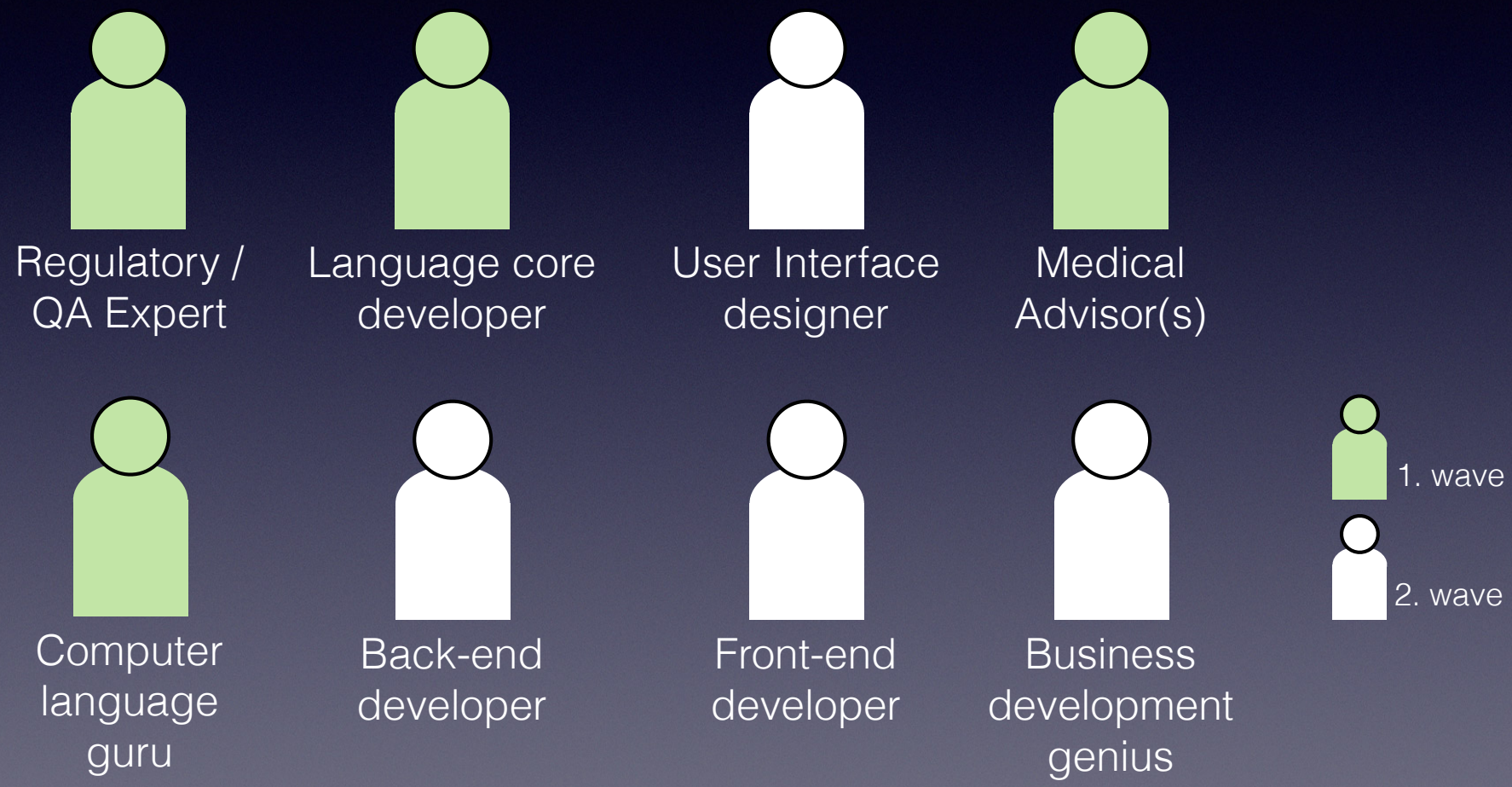




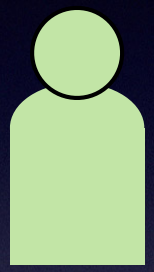
Stepwise approach to develop Hippocrates core

	Minimal prototype	Prototype	Beta version
Base language			
Statistics			
Security			
Periods			
Events			
Values			
Value groups			
Drugs			

Ideal team composition

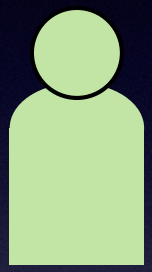


Small group to start with



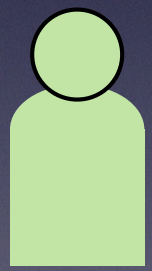
Regulatory / QA Expert

Ensure regulatory compliance of overall solution (should have strong relationship with regulatory bodies)



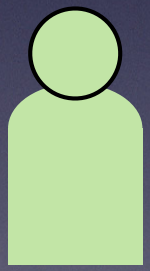
Language core developer

Develop the platform independent language core



Computer language guru

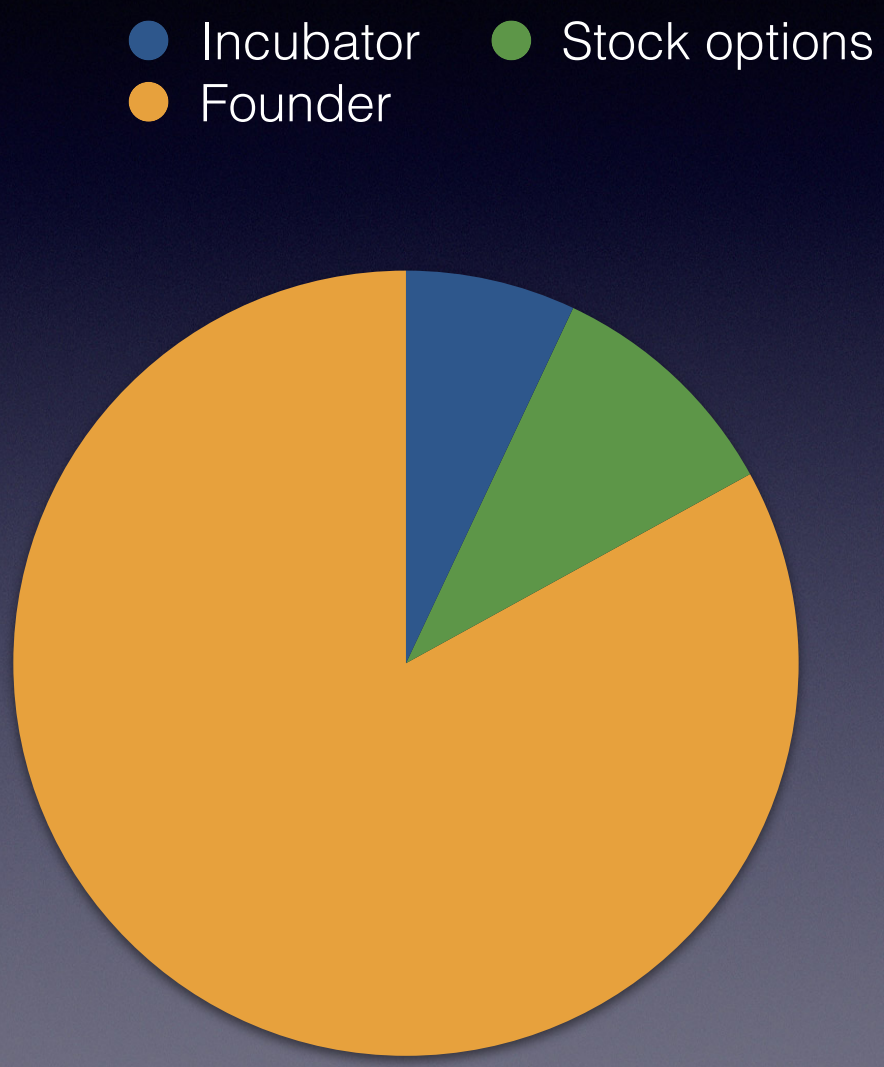
Help to create the language specification (could also be a developer)



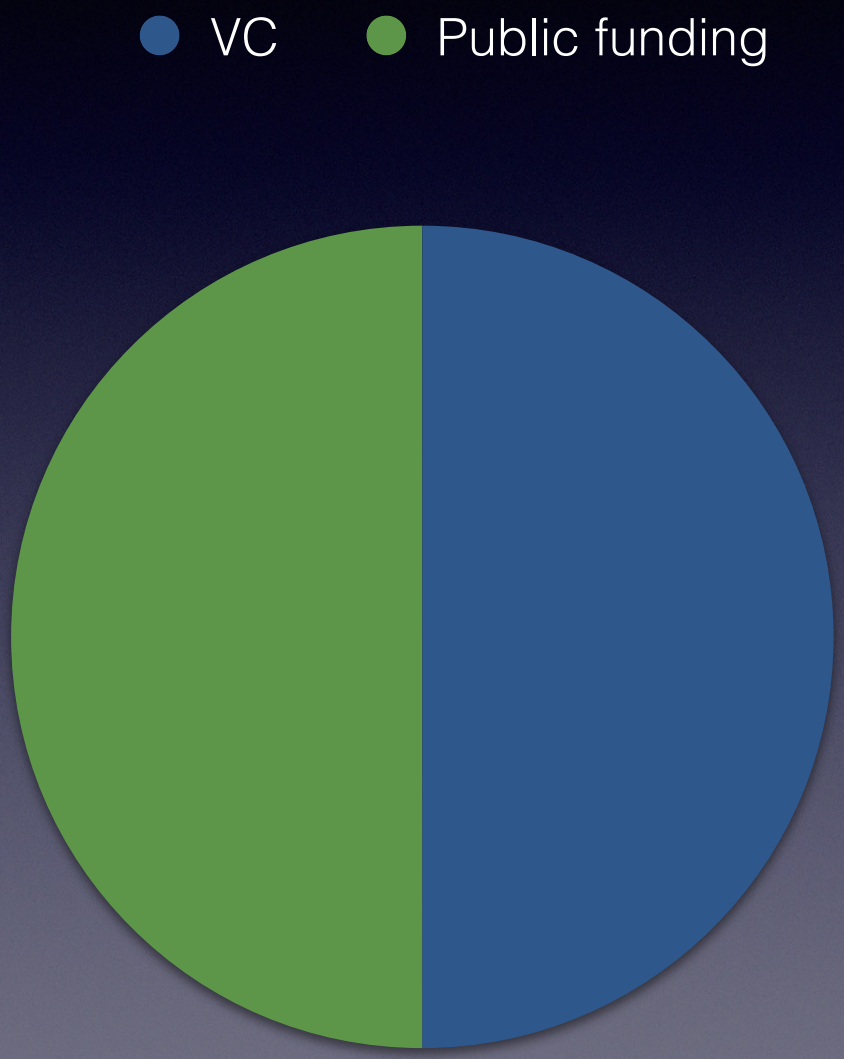
Medical Advisor(s)

Support the penetration of the medical community form the very beginning (must not be part of the actual core team)

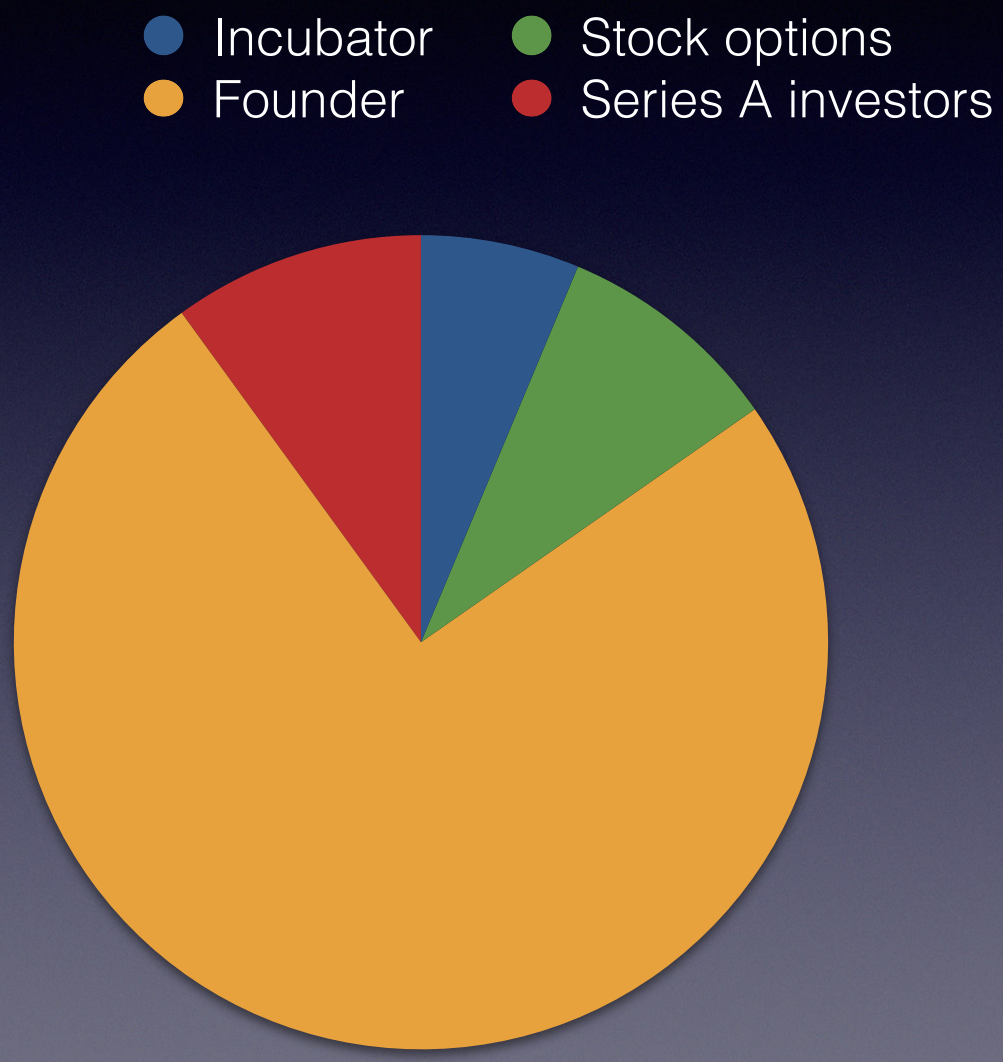
Initial structure



Funding split for Series A



Structure after Series A



Potential investors

PracticeFusion

- Kleiner Perkins Caufield & Byers
- Artis Ventures
- Qualcomm Ventures
- Orbimed Advisors
- Deerfield Management Company

HealthTap

- Great Oaks Venture Capital (seed)
- Mohr Davidow Ventures (seed & follow on)
- Mayfield Fund
- Khosla Ventures
- Asset Management Ventures

ZocDoc

- Bezos Expeditions
- DST Global
- Founders Fund
- Goldman Sachs
- Khosla Ventures
- Marc Benioff
- SV Angel

Underlined companies might have highest industry expertise

FAQ

- Does Hippocrates replace EHR systems?
  - No, Hippocrates is mainly intended to improve the interaction between the health care professional and the patient.
- Can everybody create Hippocrates plans?
  - Yes & no, Hippocrates is publicly available language. For the moment it is planned that in order in create executable plans a key will be required that will only be provided to registered professionals.
- In which geographical regions is Hippocrates available?
  - Hippocrates has no geographical limitations.

- We have Apple HealthKit and Google Fit. Why do we need Hippocrates?
  - Both are mainly intended to provide storage for health data. Hippocrates can leverage this storage. However, Hippocrates will truly enable the actual care process and provides means for health care professionals to innovate without being a software developer.
-

Support of all fields

Prevention health and wellness	Diagnosis	Intervention and therapy	Treatment, monitoring and follow-up
Hippocrates	Hippocrates	Hippocrates	Hippocrates

Easy reading

Arden*

```
IF age < 3 months  THEN
  K_Lower_Limit := 4.0; /* mEq/L */
ELSEIF age < 1 year THEN
  K_Lower_Limit := 3.5;
ELSEIF age < 60 years THEN
  K_Lower_Limit := 3.6;
ELSE /* age older than 60 years */
  K_Lower_Limit := 3.9;
ENDIF
```

- "classic" software development style

Hippocrates

```
assess age (
  0 years (
    assess pediatric age (
      0 months ... 2 months  (K Lower Limit = 4.0 mEq/L.).
      3 months ... 12 months (K Lower Limit = 3.5 mEq/L.).
  1 year ... 59 years        (K Lower Limit = 3.6 mEq/L.).
  in all other cases         (K Lower Limit = 3.9 mEq/L.).
).
```

- Easy to read ranges
- Unit are always next to numbers

* Source: The Pharmacy Informatics Primer by Doina Dumitru (Sep 1, 2008)

Experience that went into the concept

30 years of Software Development

5 years top tier strategy consulting

6 years in genome analytics (own company)

4 years in Pharma

Hippocrates

Risks

Potential risk	Impact	Mitigation(s)
Uptake in medical community too slow	Getting surpassed by other technologies / companies	1. Partner very early on with academia 2. Create incentive system for potential multipliers (a.) academia; b.) EHR provider; ...) 3. Transfer Hippocrates concept to other areas like home automation
Regulatory concerns from potential users	Slow/no uptake	1.) Staff team very early on with regulatory experts 2.) Establish advisory board with relationship to regulatory bodies (like FDA).